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The Role of Green Deal in Strengthening the Energy Security of the European Union from 2019 to 2024

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Relevance

Academic Relevance:

- Builds on existing debates around energy security, decarbonisation, and EU strategic autonomy.
- Contributes to the integration of energy policy and foreign policy perspectives.
- Engages with competing definitions of energy security in the context of climate goals.

Non-Academic Relevance:

- **Post-2019 crisis context:** EU facing multiple shocks — COVID-19, energy price surge, special military operation in Ukraine
- **Strategic urgency:** Reducing fossil fuel dependency becomes geopolitical priority
- **Green Deal shift:** From climate strategy → tool for energy security



Literature Review

Source	Contribution
Winzer (2012)	Provides a multidimensional conceptualisation of energy security, highlighting the challenge of a unified definition.
Ang et al. (2015)	Clarifies various dimensions and indicators of energy security, offering a foundation for quantitative assessment.
Goldthau & Sovacool (2012)	Argues that energy security is inseparable from justice and governance, framing it as a unique policy problem.
Cherp & Jewell (2011, 2014)	Introduce sovereignty, robustness, and resilience as competing logics in energy security; advocate for an integrated framework.
Christou & Adamides (2013)	Applies securitisation theory to energy, showing how climate and energy discourses evolve in response to crises.
Pianta & Lucchese (2020)	Analyses the Green Deal as an industrial policy for a just transition, focusing on equity and social cohesion.
Schittekatte et al. (2020)	Critically assesses TEN-E regulation's compatibility with Green Deal goals; proposes selection and funding reforms.
McGowan (2011)	Traces how EU energy insecurity responses evolved from the 1970s to 2000s, providing essential background.
Leonard et al. (2021)	Examines how the Green Deal repositions EU geopolitics; highlights climate-policy diplomacy tensions.
Hafner & Raimondi (2020); Miçek et al. (2022); Tutak et al. (2021)	Focus on policy and decarbonisation metrics but insufficiently explore strategic security or integration aspects.
Hainsch et al. (2022)	Use modelling and scenario analysis, lacking real-world political and infrastructural developments.



Research Gap

→ **Conceptual frameworks dominate**

Most studies (e.g., Cherp & Jewell, 2011; Winzer, 2012) focus on defining energy security but lack empirical testing of the Green Deal's instruments.

→ **Security effects of the Green Deal are understudied**

Literature reviews the Green Deal as a climate or industrial strategy (e.g., Pianta & Lucchese, 2020), but few assess its impact on strategic energy vulnerability.

→ **Geopolitical disruptions are insufficiently integrated**

The 2022 energy crisis and its implications for EU energy security are not fully explored in major studies (e.g., Leonard et al., 2021 focus on strategy, not outcomes).

→ **Underexplored convergence of climate and security discourse**

Although securitisation theory is well-established, there is limited research on how EU political rhetoric after 2019 increasingly framed climate action as a component of energy security strategy.

→ **Insufficient analysis of policy and infrastructure integration**

The interaction between internal market regulation (e.g., TEN-E) and external energy diplomacy (e.g., REPowerEU) is rarely assessed in terms of their contribution to long-term energy resilience.



Research Question

How does the European Green Deal contribute to enhancing the energy security of the EU from 2019 to 2024?

Hypothesis

The European Green Deal has strengthened the energy security of the EU by promoting diversification of energy sources and reducing dependency on external fossil fuels.

Expected Outcomes

- European Green Deal has contributed to reducing the EU's strategic energy vulnerability by diversifying supply sources, promoting renewables, and deepening institutional integration
- Emerging challenges, including new dependencies on critical materials, social inequities, and geopolitical risks, that may undermine long-term energy security

Subject & Object

The **subject** of the study is the transformation of EU energy security through climate policy

The **object** is the European Green Deal and its energy-related measures between 2019 and 2024

Timeframe

2019-2024

2019 – launch of the EU Green Deal

Also includes external shocks like COVID-19 and geopolitical conflicts



Methodology

Methodological approach	Mixed-method: quantitative + qualitative
Quantitative methods	Analysis of fuel import data, RES share, storage levels
Qualitative methods	Discourse analysis of EU speeches, summit declarations
Comparative Focus	Pre-/post-Green Deal (before 2019 vs 2019–2024)
Case Studies	Germany, Italy, Poland, Baltic States
Document Analysis	Legislative texts, EU communications (e.g. REPowerEU, Fit for 55, RED III)
Limitations	Timeframe limits long-term conclusions; limited internal data; rhetoric bias



Theoretical Framework

- **Complex Interdependence** (Keohane & Nye): explains how the EU enhances security through dense economic and regulatory ties, reducing vulnerability by diversifying partners and deepening cooperation.
- **Securitisation Theory** (Buzan & Waever): used to interpret how EU leaders frame energy as a security issue requiring extraordinary policy responses, justifying tools like joint gas purchasing.
- **Strategic Autonomy**: focuses on the EU's ability to act independently in key areas such as critical raw materials, hydrogen, and carbon policy, linking climate targets to geopolitical capacity (Biscop & Tocci)
- **Energy Justice**: expands the analysis to assess fairness, equity, and social outcomes of energy transition instruments like the Just Transition Mechanism and Social Climate Fund (Sovacool & McCauley).



Structure of the Work

Introduction

Chapter 1: Historical Context of EU Energy Security from 1990s to 2019

1.1 Evolution of EU Energy Security from 1990s to 2010s

1.2 Structural and Geopolitical Challenges in European Energy Policy Before 2019

Chapter 2: The European Green Deal’s Impact on EU Energy Security from 2019 to 2024

2.1 Policy Architecture and Instrumentation under the European Green Deal

2.2 Empirical Outcomes and Comparative Assessment of Energy-Security Indicators

2.3 Crisis-Induced Adaptation and the Securitization of EU Energy Policy

Chapter 3: Scenarios, Risks and Strategic Options for EU Energy Security beyond 2024

3.1 Risk Landscape and Unintended Consequences from 2024 to 2035

3.2 Forward Scenarios to 2035

3.3 Policy Tools for a Resilient and Secure System

Conclusion

Reference List



1.1 Evolution of EU Energy Security from 1990s to 2010s

- The EU's energy policy emerged from economic integration, but security concerns began to influence decisions after the 1970s oil crises and the gas disputes of the early 2000s.
- In the 1990s, the focus was on market liberalisation, creating a single energy market to improve efficiency and competitiveness.
- Energy was not formally part of the EU's core competencies until the Treaty of Lisbon (2009), which for the first time established a legal basis for a common energy policy.
- The 2006 and 2009 gas crises with Russia revealed the vulnerability of Eastern and Central European countries, pushing energy security higher on the agenda.
- The EU responded by developing the **first Energy Packages** and launching the **Trans-European Energy Networks (TEN-E)** to improve infrastructure and interconnectivity.

→ These early developments laid the groundwork for later securitisation of energy and revealed the EU's growing dependence on external suppliers, especially Russia.



1.2 Structural and Geopolitical Challenges in European Energy Policy Before 2019

- The 2006 and 2009 Russia-Ukraine gas disputes disrupted supplies to several EU countries, demonstrating the geopolitical risks of energy dependency.
- These crises prompted the EU to shift from purely economic goals to incorporating **strategic and security concerns** into energy policy.
- Key institutional responses included the **Second and Third Energy Packages**, which aimed to improve market transparency, unbundle supply and transmission, and enhance consumer protection.
- **European Energy Security Strategy (2014)** → framed energy supply as a strategic vulnerability and proposed diversification, storage, efficiency, and solidarity mechanisms.
- The EU also began developing reverse-flow infrastructure and LNG terminals to reduce dependency on Russian pipeline gas.

→ These years solidified the security dimension of energy policy and laid the foundation for treating energy as a geopolitical asset.



2.1 Policy Architecture and Instrumentation under the European Green Deal

The Green Deal reconceptualises energy policy through a climate-first lens, reframing decarbonisation as a security imperative.

Key legislative packages, such as *Fit for 55*, *REPowerEU*, and *RED III*, aim not only at climate neutrality but also at structural resilience.

Three energy-security pillars:

- **Decarbonisation:** Reduced fossil-fuel imports via accelerated rollout of renewables and clean hydrogen.
- **Integration:** Expansion of cross-border infrastructure and common market mechanisms
- **Resilience:** Boosted storage, diversification of supply, and crisis-readiness, particularly after 2022 disruptions.

It also shows how climate policy became a strategic response to geopolitical risks, supported by regulatory innovation and market coordination.



2.2 Empirical Outcomes and Comparative Assessment of Energy-Security Indicators

Russian gas, which accounted for **45% of EU gas imports in 2021** (~155 bcm), dropped below **20% by the end of 2022**, following REPowerEU efforts and emergency LNG diversification measures.

Russian oil imports, formerly **25–30% of EU oil supply**, were phased out due to sanctions, and coal imports fell from **nearly 50% to zero** by late 2022 due to embargo measures.

In parallel, **solar and wind capacity in the EU rose by 60% between 2019 and 2023**, reaching record levels in 2023 with **over 60 GW of new additions** (IEA, 2024).

Gas storage fill rates exceeded **95% in November 2023**, improving resilience ahead of winter and signaling stronger coordinated infrastructure usage (GIE, 2023).

Instruments like **REPowerEU**, the **Hydrogen Strategy**, and **RED III** provided legislative and financial frameworks to accelerate this energy shift.

→ The EU significantly reduced its fossil-fuel exposure to Russia, but new dependencies emerged—such as **increased LNG imports from the U.S. and Qatar**, and reliance on **critical raw materials** for renewables. **Thus, strategic vulnerability was reshaped, not eliminated.**



2.3 Crisis-Induced Adaptation and the Securitization of EU Energy Policy

Political language escalated from “market diversification” (2015–2019) to “energy sovereignty” (2019–2021) and eventually to “energy weaponization” in 2022.

The EU adopted extraordinary measures such as **joint gas purchasing**, emergency permitting for renewables, and **price caps** under a new “crisis framework.”

Leaders increasingly framed climate neutrality as a **security imperative**, not just an environmental goal, fusing the Green Deal with geopolitical priorities.

The rapid uptake of emergency tools (e.g., **the Gas Storage Regulation (2022)** and **Energy Platform Task Force**) reflected a securitized view of energy resilience.

The **temporary coal revival** in 2022 was rhetorically justified as a safeguard while simultaneously accelerating RES targets for 2030 and beyond.

→The securitisation of energy policy allowed for exceptional interventions that would have been politically unfeasible under normal conditions. This shift demonstrates how crises were leveraged to deepen integration and push the Green Deal as a security strategy.



3.1 Risk Landscape and Unintended Consequences from 2024 to 2035

- This subchapter outlines emerging risks that may undermine the long-term security benefits of the Green Deal beyond 2024.
- While fossil fuel dependence declined, **new vulnerabilities** have arisen:
 - **Increased reliance on LNG**, particularly from the U.S. and Qatar, creates geopolitical and price-exposure risks.
 - **Critical raw materials** like lithium, rare earths, and cobalt are now essential for the energy transition, but the EU depends heavily on imports (e.g., 97% of magnesium from China; EC, 2023a).
- The **expansion of hydrogen infrastructure** and green technologies brings industrial gains, but also poses risks of technological lock-in and unequal development between Member States.
- The **EU's new dependencies shift from hydrocarbons to supply chains** and infrastructure controlled by third parties, particularly in clean tech and strategic minerals.
- Social risks also persist: unequal cost distribution may trigger **justice-related backlash** in low-income or coal-dependent regions despite mechanisms like the Social Climate Fund.
- → The Green Deal reduces traditional energy dependencies but generates new strategic and socio-economic vulnerabilities. Addressing these is essential to ensure long-term, just, and resilient energy security.



3.2 Forward Scenarios to 2035

Scenario	Description	Energy Security Outlook
Accelerated Green Deal	Full implementation of Fit-for-55 and REPowerEU targets, massive RES and hydrogen deployment, strong investment in grids and interconnectors.	High resilience, low fossil import dependency, increased autonomy and flexibility.
Status Quo Extension	Moderate renewables growth with patchy national implementation, partial success on hydrogen, slow infrastructure upgrades.	Mixed results, vulnerabilities persist in some sectors and regions.
Crisis Reversal	Return to fossil fuels due to repeated crises, delays in renewables, grid bottlenecks, and limited green investment.	High vulnerability, increased LNG dependence, setback in climate-security linkage.
Geopolitical Fragmentation	Divergence in national energy policies, collapse of coordinated action, external supplier dependency re-emerges.	Fragmented security, weak coordination, strategic exposure to external shocks.



3.3 Policy Tools for a Resilient and Secure System

Strategic Autonomy Tools

- *Critical Raw Materials Act* – secures access to essential inputs for clean energy technologies
- *Carbon Border Adjustment Mechanism* – protects EU industries from carbon leakage and external shocks

Social Cohesion & Fairness

- *Just Transition Mechanism* – supports regions most affected by the green shift
- *Social Climate Fund* – mitigates social and regional inequalities linked to energy reform

Institutional Integration

- *REPowerEU* – accelerates infrastructure projects and diversifies supply
- *Joint gas purchasing and Hydrogen Alliances* – reinforce collective EU-level energy strategy

Conclusion

- The Green Deal's security dimension rests on a combination of geopolitical autonomy, environmental ambition, and social legitimacy
- Success depends on fast implementation, sufficient funding, and strong Member State cooperation



Conclusion

The Green Deal reshaped the EU's energy policy by integrating climate and security goals into a single strategic vision. It redefined energy security beyond fossil fuel diversification toward systemic resilience and clean autonomy.

All four analytical lenses – complex interdependence, securitisation theory, strategic autonomy, and energy justice – confirmed that the Green Deal functions as both a **climate** and a **security** instrument.

Empirical evidence from the chapters showed:

- a discursive shift in EU language post-2019 from "decarbonisation" to "sovereignty";
- material responses like joint gas procurement and accelerated renewables;
- institutional innovations like the Social Climate Fund and Critical Raw Materials Act.

Short-term fossil backtracks (e.g. coal reopenings) do not contradict the Green Deal's trajectory but reflect **security-driven temporal trade-offs**.

The research confirms the hypothesis:

The European Green Deal, initially framed as an environmental strategy, has become a core component of the EU's energy security architecture.

The hypothesis has been proven

